317521: Computer Networks

Unit Number : 6
Unit Name : Medium Access Control
Unit Outcomes : CO6
CO6 : Summarize concepts of MAC and ethernet

Syllabus

- Channel allocation: Static and Dynamic, Multiple Access Protocols: Pure and Slotted ALOHA, CSMA, WDMA
- IEEE 802.3 Standards and Frame Formats
- CSMA/CD, Binary Exponential Back -off algorithm
- Fast Ethernet, Gigabit Ethernet, IEEE 802.11a/b/g/n and IEEE 802.15 and IEEE 802.16 Standards, Frame formats, CSMA/CA.

Data Link Layer

The data link layer is used in a computer network to transmit the data between two devices or nodes. It divides the layer into parts such as **data link control** and the **multiple access resolution/protocol**. The upper layer has the responsibility to flow control and the error control in the data link layer, and hence it is termed as **logical of data link control**. Whereas the lower sub-layer is used to handle and reduce the collision or multiple access on a channel. Hence it is termed as Medium Access Control.

Medium Access Control:

A sublayer of the Data Link Layer (Layer 2) in the OSI (Open Systems Interconnection) model of computer networking is called Medium Access Control (MAC). Its main duty is to control access to a network's shared communication medium, ensuring that various devices can transmit data effectively and fairly over a common communication path like a wired Ethernet or wireless Wi-Fi network.

Function of Medium Access Control

- A computer network's Medium Access Control (MAC) layer performs several crucial tasks to control access to the shared communication medium.
- Its main responsibility is preventing collisions while ensuring multiple devices can transmit data fairly and efficiently over a shared communication channel.

The MAC layer performs the following primary tasks:

•**Access Control:** The MAC layer controls which device can transmit data at any given time by controlling access to the shared communication medium. It employs various access control techniques to control how devices compete for access to the medium. Contention-based (like Carrier Sense Multiple Access with Collision Detection or CSMA/CD) and contention-free (like token passing) techniques can be used.

•**Frame Addressing:** Networked devices on the same network segment are uniquely identified by their MAC addresses, also called hardware or physical addresses. The MAC layer includes the source and destination MAC addresses in data frames to identify the sender and recipient of the data.

•**Frame Formatting:** The MAC layer packages data from the higher layers (typically the Network Layer) into frames that can be transmitted over the network medium. These frames contain data, control information, information for checking for errors, and information for addressing.

•**Error detection:** Many MAC protocols include error-checking components to identify transmission errors. This guarantees the accuracy of the data transmitted through the medium. The MAC layer may ask for retransmission of the frame if a mistake is found.

•**Frame detection and collision handling:** Collisions can happen when multiple devices try to transmit data simultaneously over a shared communication medium like Ethernet. Detecting collisions and putting collision resolution mechanisms into place to lessen their effects falls to the MAC layer. For instance, when CSMA/CD detects collisions, it starts a back-off mechanism that sends data again after an arbitrary amount of time has passed.

•**Flow Control:** Some MAC protocols employ flow control to ensure that data is transmitted at a rate the recipient device can handle without experiencing data loss or overflow. Flow control mechanisms may use feedback from the receiver to the sender to change transmission rates.

•**Address Resolution:** In Ethernet networks, the MAC layer uses the Address Resolution Protocol (ARP) to translate addresses from higher layers (like IP addresses) to MAC addresses. The local network segment's corresponding MAC address is found using ARP, which maps the destination IP address to it.

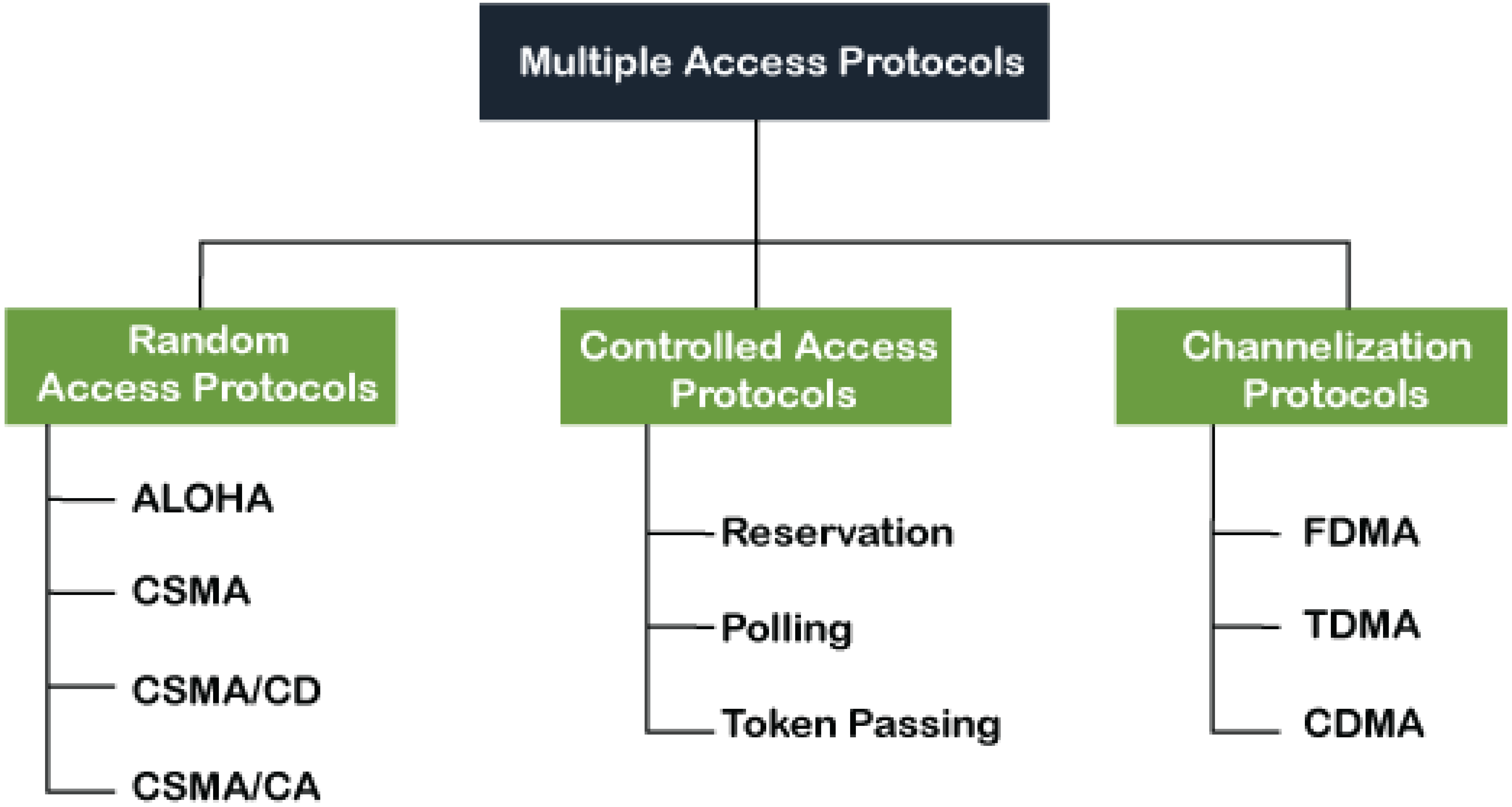
•**Broadcast and Multicast:** The MAC layer supports both broadcasting and multicasting, allowing for the sending of frames to various groups of devices (multicast) or all devices on a network segment (broadcast) as needed.

•**Security:** Some MAC layer protocols include security features like encryption and authentication to protect data and ensure that only authorized devices can access the network.

What is a multiple access protocol:

When a sender and receiver have a dedicated link to transmit data packets, the data link control is enough to handle the channel. Suppose there is no dedicated path to communicate or transfer the data between two devices. In that case, multiple stations access the channel and simultaneously transmits the data over the channel. It may create collision and cross talk. Hence, the multiple access protocol is required to reduce the collision and avoid crosstalk between the channels.

Following are the types of multiple access protocol that is subdivided into the different process as:



A. Random Access Protocol

In this protocol, all the station has the equal priority to send the data over a channel. In random access protocol, one or more stations cannot depend on another station nor any station control another station. Depending on the channel's state (idle or busy), each station transmits the data frame. However, if more than one station sends the data over a channel, there may be a collision or data conflict. Due to the collision, the data frame packets may be lost or changed. And hence, it does not receive by the receiver end.

Following are the different methods of random-access protocols for broadcasting frames on the channel.

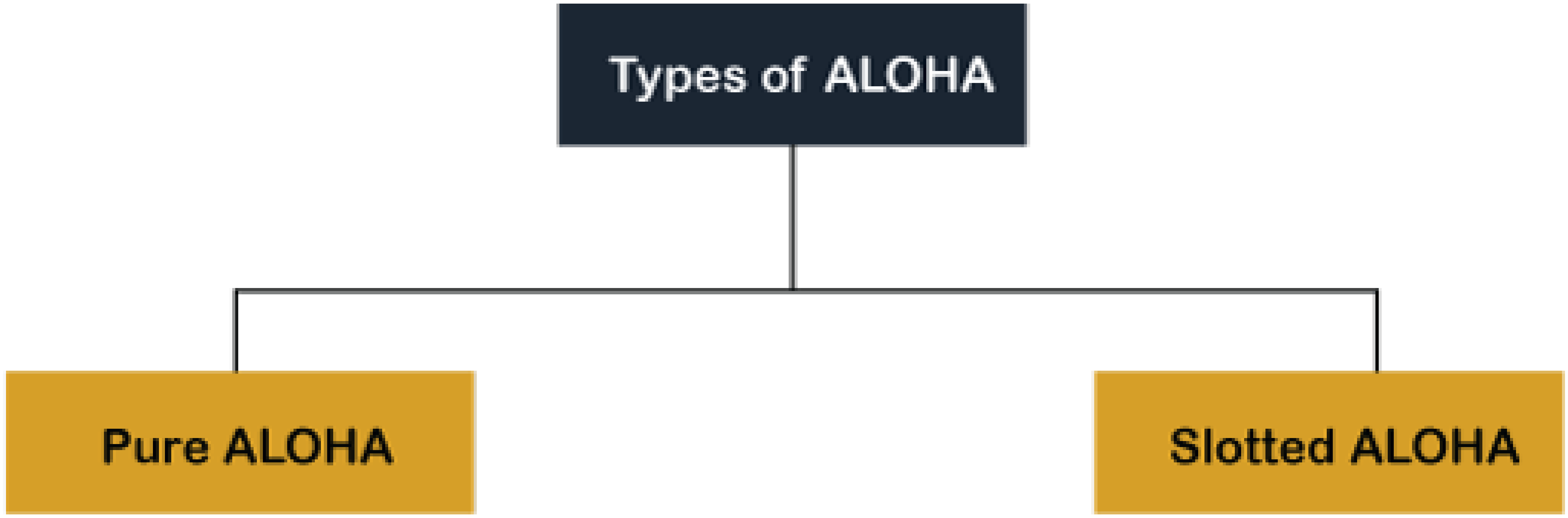
- Aloha
- CSMA
- CSMA/CD
- CSMA/CA

ALOHA Random Access Protocol

It is designed for wireless LAN (Local Area Network) but can also be used in a shared medium to transmit data. Using this method, any station can transmit data across a network simultaneously when a data frameset is available for transmission.

Aloha Rules

- Any station can transmit data to a channel at any time.
- It does not require any carrier sensing.
- Collision and data frames may be lost during the transmission of data through multiple stations.
- Acknowledgment of the frames exists in Aloha. Hence, there is no collision detection.
- It requires retransmission of data after some random amount of time.



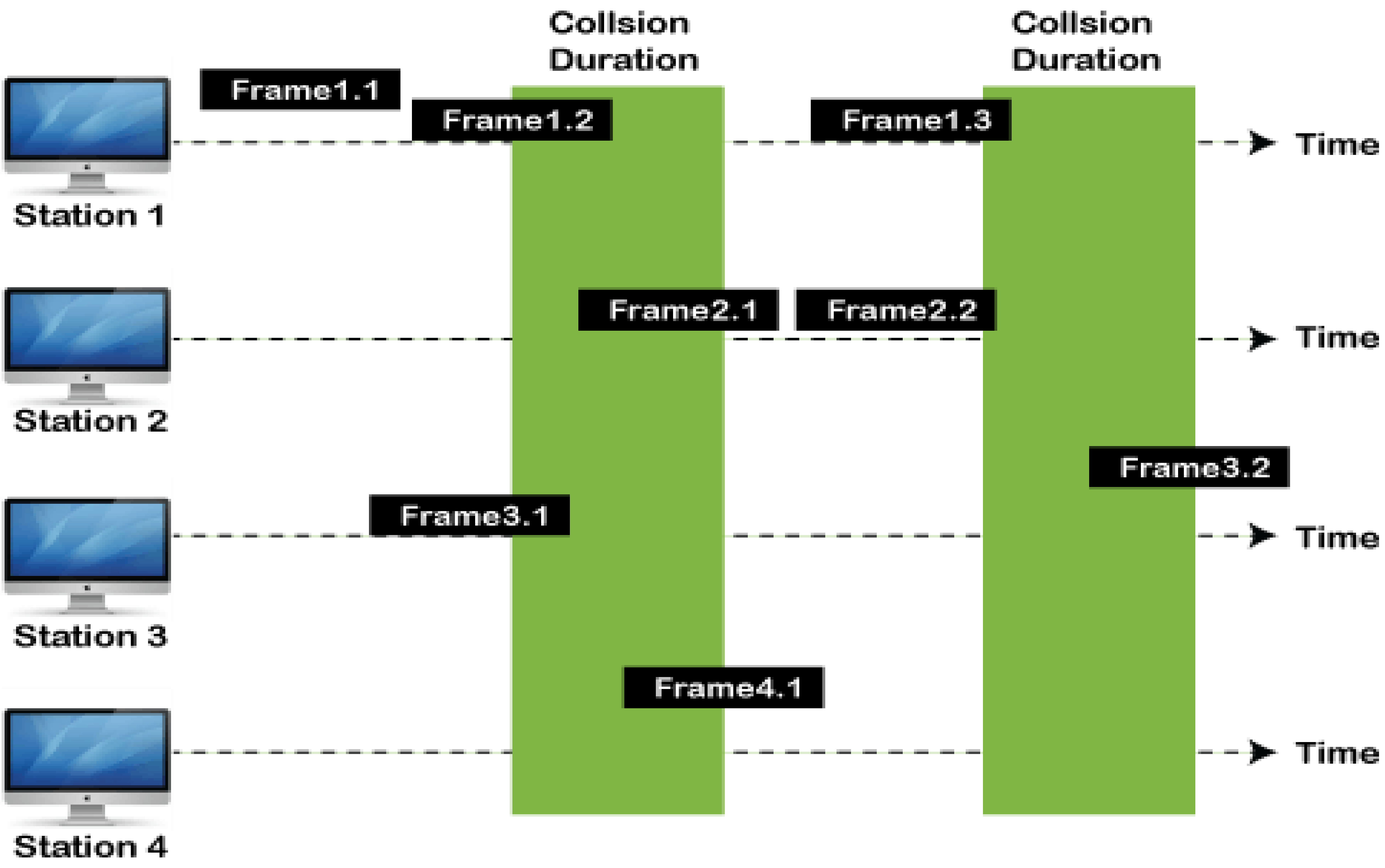
Pure Aloha

Whenever data is available for sending over a channel at stations, we use Pure Aloha. In pure Aloha, when each station transmits data to a channel without checking whether the channel is idle or not, the chances of collision may occur, and the data frame can be lost. When any station transmits the data frame to a channel, the pure Aloha waits for the receiver's acknowledgment. If it does not acknowledge the receiver end within the specified time, the station waits for a random amount of time, called the backoff time (Tb). And the station may assume the frame has been lost or destroyed. Therefore, it retransmits the frame until all the data are successfully transmitted to the receiver.

The total vulnerable time of pure Aloha is 2 * Tfr.

Maximum throughput occurs when G = 1/ 2 that is 18.4%.

Successful transmission of data frame is S = G * e ^ - 2 G.



Frames in Pure ALOHA

As we can see in the figure above, there are four stations for accessing a shared channel and transmitting data frames. Some frames collide because most stations send their frames at the same time. Only two frames, frame 1.1 and frame 2.2, are successfully transmitted to the receiver end. At the same time, other frames are lost or destroyed. Whenever two frames fall on a shared channel simultaneously, collisions can occur, and both will suffer damage. If the new frame's first bit enters the channel before finishing the last bit of the second frame. Both frames are completely finished, and both stations must retransmit the data frame.

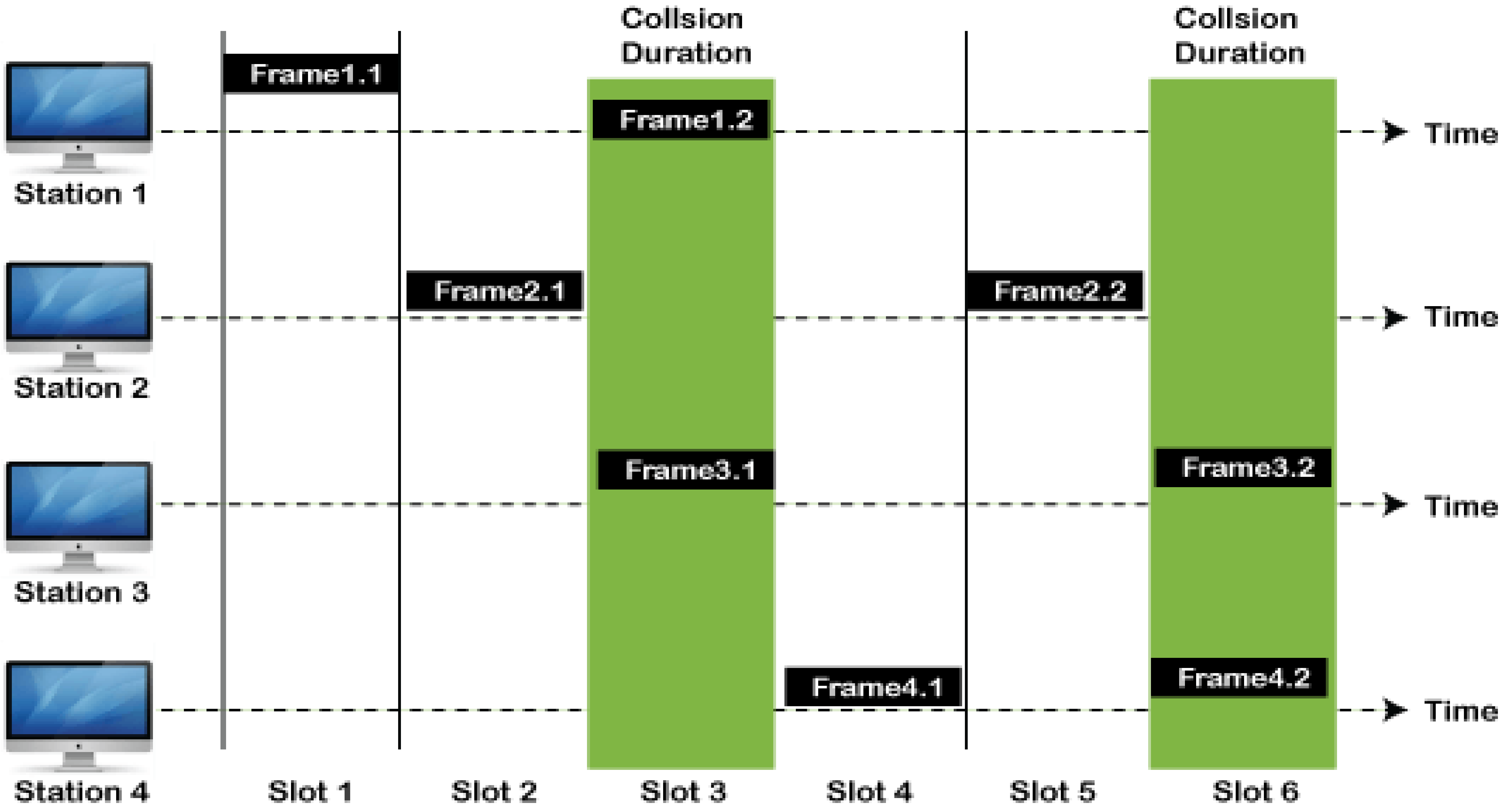
Slotted Aloha

The slotted Aloha is designed to overcome the pure Aloha's efficiency because pure Aloha has a very high possibility of frame hitting. In slotted Aloha, the shared channel is divided into a fixed time interval called **slots**. So that, if a station wants to send a frame to a shared channel, the frame can only be sent at the beginning of the slot, and only one frame is allowed to be sent to each slot. And if the stations are unable to send data to the beginning of the slot, the station will have to wait until the beginning of the slot for the next time. However, the possibility of a collision remains when trying to send a frame at the beginning of two or more station time slot.

Maximum throughput occurs in the slotted Aloha when G = 1 that is 37%.

The probability of successfully transmitting the data frame in the slotted Aloha is S = G * e ^ - 2 G.

The total vulnerable time required in slotted Aloha is Tfr.



Frames in Slotted ALOHA

CSMA (Carrier Sense Multiple Access)

It is a **carrier sense multiple access** based on media access protocol to sense the traffic on a channel (idle or busy) before transmitting the data. It means that if the channel is idle, the station can send data to the channel. Otherwise, it must wait until the channel becomes idle. Hence, it reduces the chances of a collision on a transmission medium.

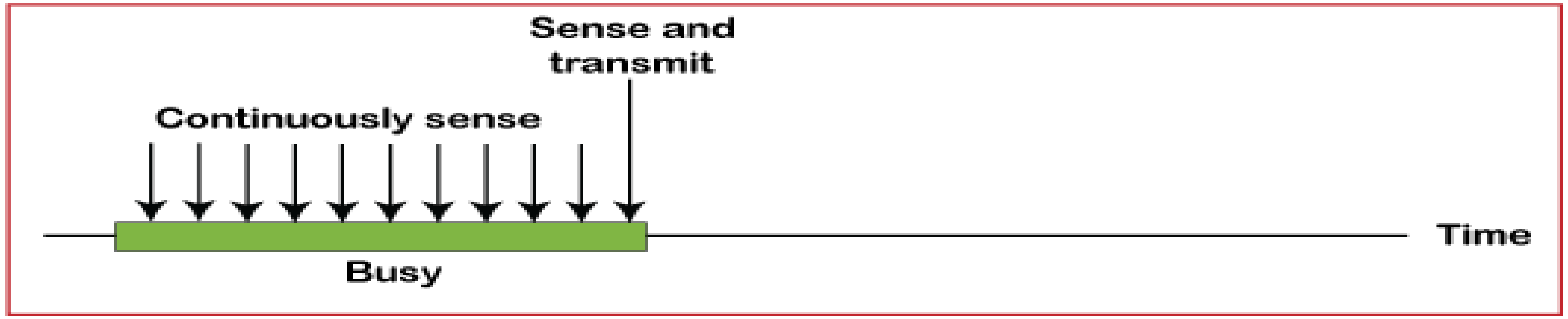
CSMA Access Modes

1-Persistent: In the 1-Persistent mode of CSMA that defines each node, first sense the shared channel and if the channel is idle, it immediately sends the data. Else it must wait and keep track of the status of the channel to be idle and broadcast the frame unconditionally as soon as the channel is idle.

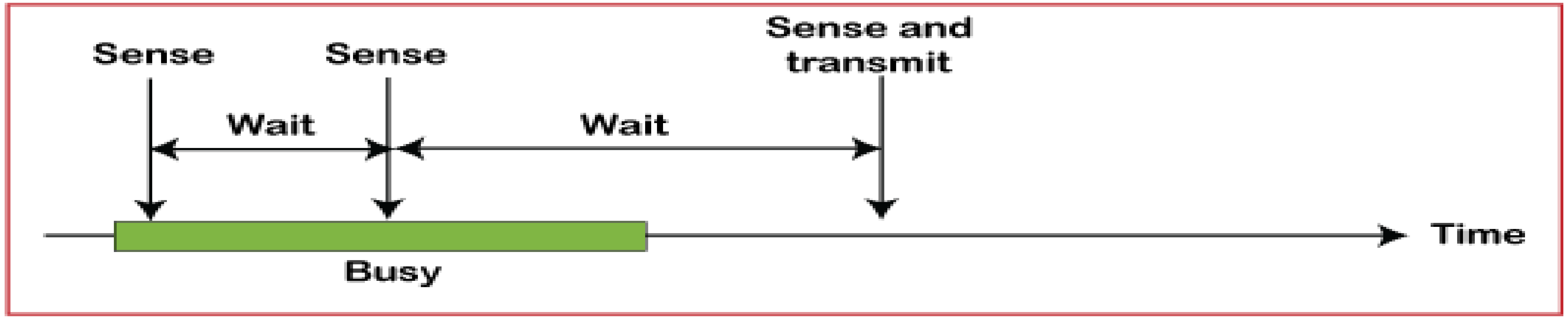
Non-Persistent: It is the access mode of CSMA that defines before transmitting the data, each node must sense the channel, and if the channel is inactive, it immediately sends the data. Otherwise, the station must wait for a random time (not continuously), and when the channel is found to be idle, it transmits the frames.

P-Persistent: It is the combination of 1-Persistent and Non-persistent modes. The P-Persistent mode defines that each node senses the channel, and if the channel is inactive, it sends a frame with a **P** probability. If the data is not transmitted, it waits for a (**q = 1-p probability**) random time and resumes the frame with the next time slot.

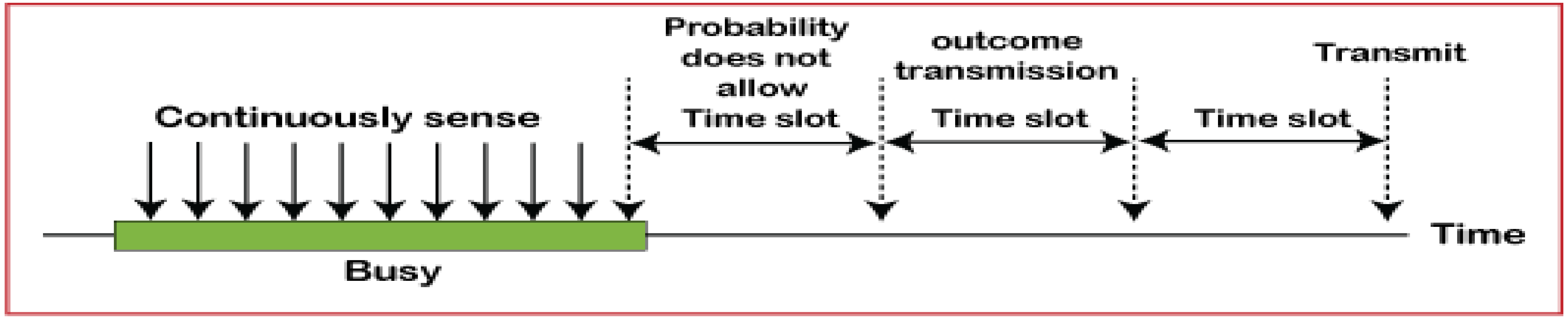
O- Persistent: It is an O-persistent method that defines the superiority of the station before the transmission of the frame on the shared channel. If it is found that the channel is inactive, each station waits for its turn to retransmit the data.



a. 1-persistent



b. Nonpersistent



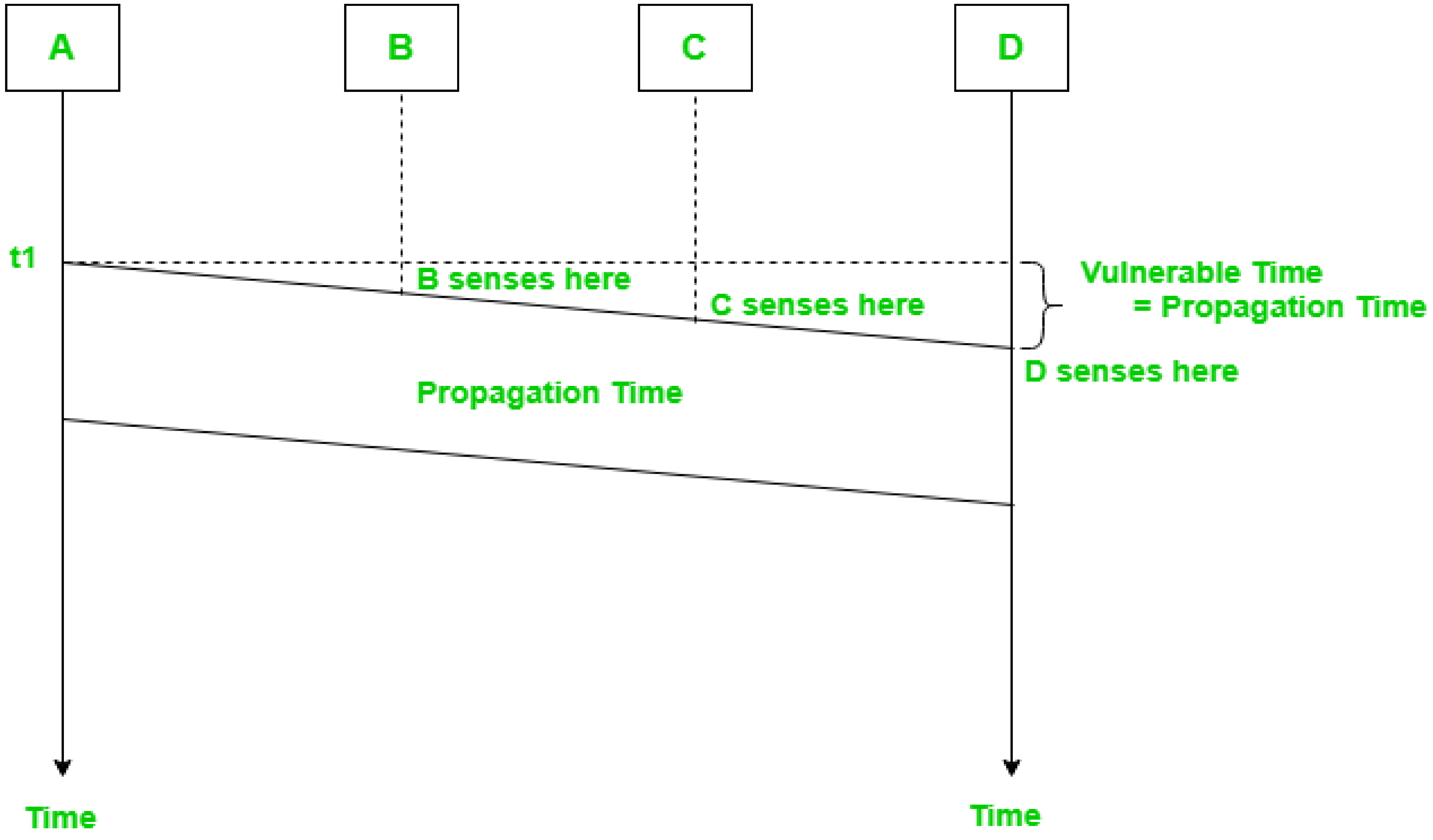
c. p-persistent

Carrier Sense Multiple Access (CSMA)

This method was developed to decrease the chances of collisions when two or more stations start sending their signals over the data link layer. Carrier Sense multiple access requires that each station **first check the state of the medium** before sending.

Vulnerable Time:

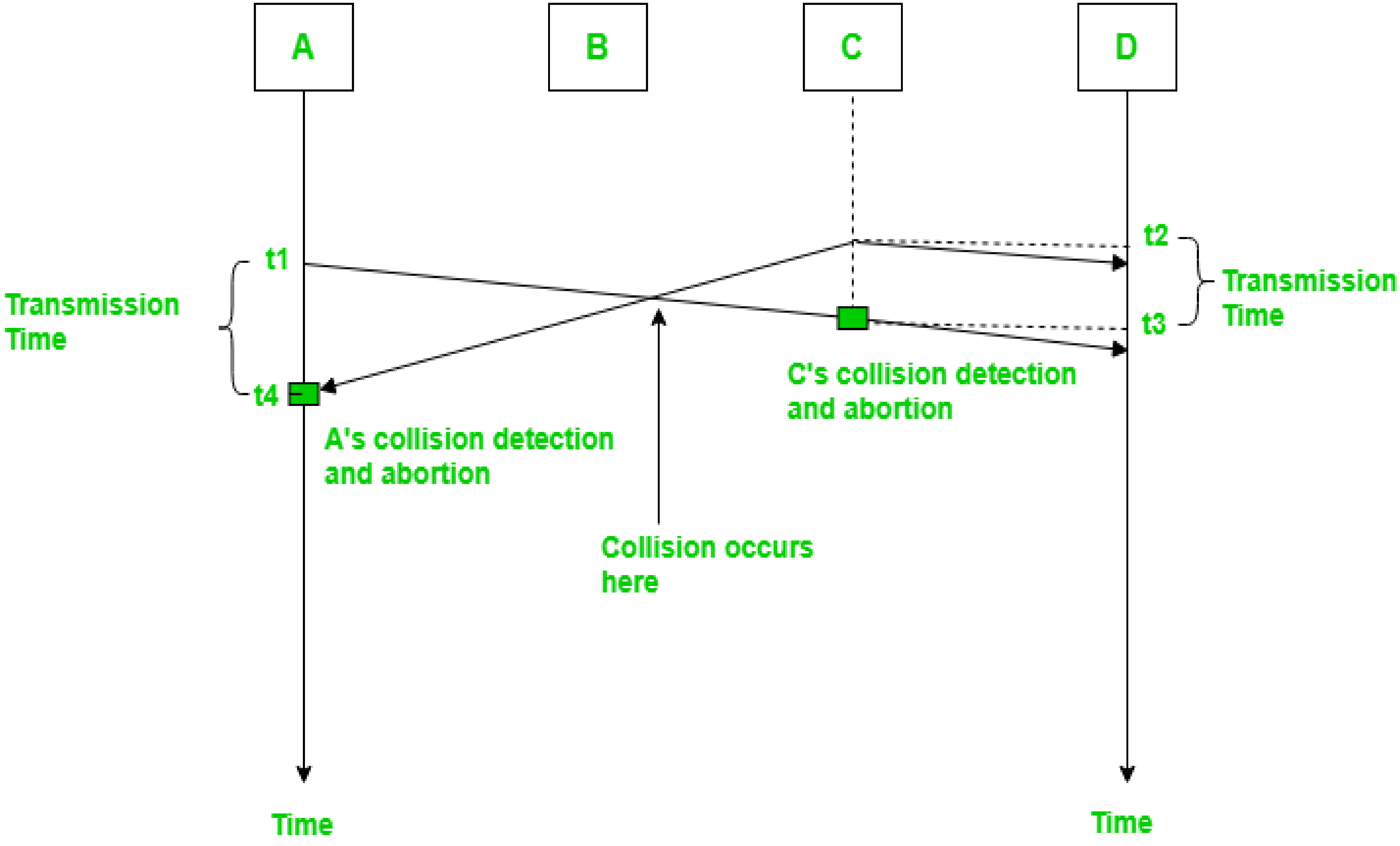
Vulnerable time = Propagation time (Tp)



The persistence methods can be applied to help the station take action when the channel is busy/idle.

1. Carrier Sense Multiple Access with Collision Detection (CSMA/CD):

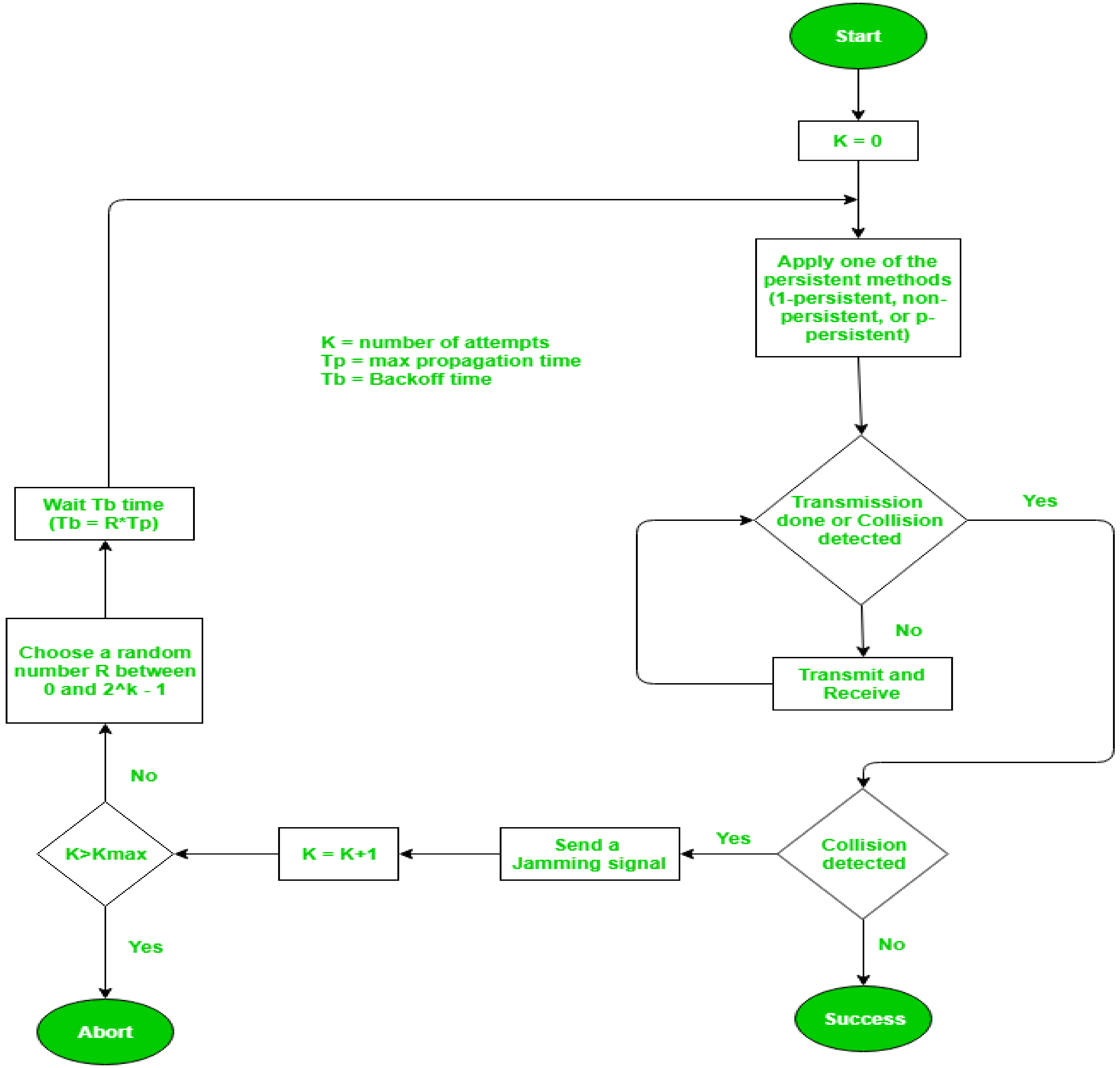
In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If successful, the transmission is finished, if not, the frame is sent again.



In the diagram, *starts* sending the first bit of its frame at t1 and since C sees the channel idle at t2, starts sending its frame at t2. C detects A's frame at t3 and aborts transmission. A detects C's frame at t4 and aborts its transmission. Transmission time for C's frame is, therefore, t3-t2 and for A's frame is t4-t1

So, the **frame transmission time (Tfr) should be at least twice the maximum propagation time (Tp)**. This can be deduced when the two stations involved in a collision are a maximum distance apart.

Process: The entire process of collision detection can be explained as follows:



Throughput and Efficiency: The throughput of CSMA/CD is much greater than pure or slotted ALOHA.

•For the 1-persistent method, throughput is 50% when G=1.

•For the non-persistent method, throughput can go up to 90%.

2. Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) –

The basic idea behind CSMA/CA is that the station should be able to receive while transmitting to detect a collision from different stations. In wired networks, if a collision has occurred then the energy of the received signal almost doubles, and the station can sense the possibility of collision. In the case of wireless networks, most of the energy is used for transmission, and the energy of the received signal increases by only 5-10% if a collision occurs. It can’t be used by the station to sense collision. Therefore **CSMA/CA has been specially designed for wireless networks**.

These are three types of strategies:

1.InterFrame Space (IFS): When a station finds the channel busy it senses the channel again, when the station finds a channel to be idle it waits for a period of time called **IFS time**. IFS can also be used to define the priority of a station or a frame. Higher the IFS lower is the priority.

2.Contention Window: It is the amount of time divided into slots. A station that is ready to send frames chooses a random number of slots as **wait time**.

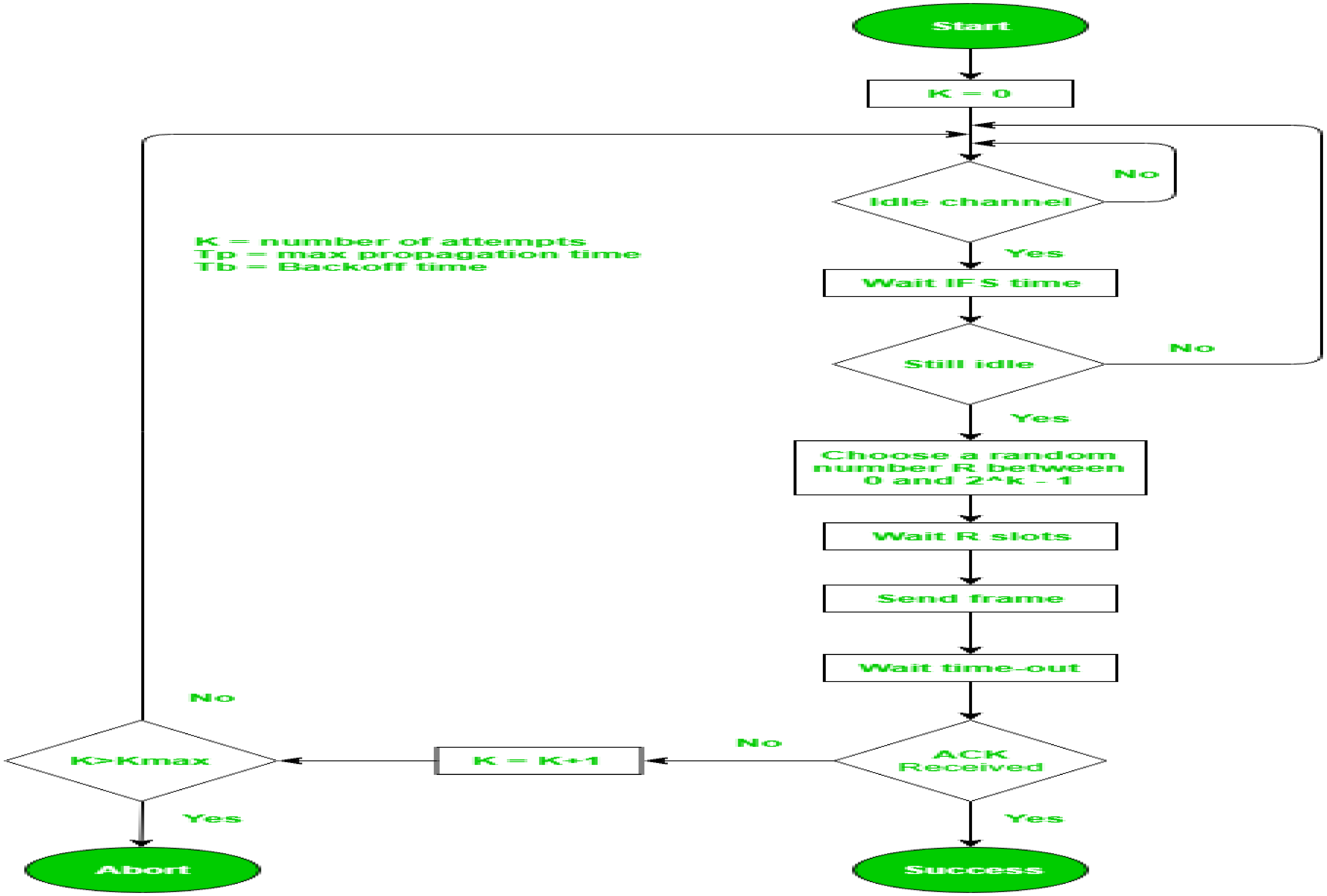
3.Acknowledgments: The positive acknowledgments and time-out timer can help guarantee a successful transmission of the frame.

Characteristics of CSMA/CA :

1. **Carrier Sense**: The device listens to the channel before transmitting, to ensure that it is not currently in use by another device.
2. **Multiple Access**: Multiple devices share the same channel and can transmit simultaneously.
3. **Collision Avoidance**: If two or more devices attempt to transmit at the same time, a collision occurs. CSMA/CA uses random backoff time intervals to avoid collisions.
4. **Acknowledgment (ACK)**: After successful transmission, the receiving device sends an ACK to confirm receipt.
5. **Fairness**: The protocol ensures that all devices have equal access to the channel and no single device monopolizes it.
6. **Binary Exponential Backoff**: If a collision occurs, the device waits for a random period of time before attempting to retransmit. The backoff time increases exponentially with each retransmission attempt.

Overall, CSMA/CA balances the need for efficient use of the shared channel with the need to avoid collisions, leading to reliable and fair communication in a wireless network.

Process: The entire process of collision avoidance can be explained as follows:


Start
K = 0
Idle channel
No
Yes
Wait IFS time
Still idle
No
Yes
Choose a random number R between 0 and 2^k - 1
Wait R slots
Send frame
Wait time-out
ACK Received
No
Yes
Success
K = K+1
K>Kmax
No
Yes
Abort
K = number of attempts
Tp = max propagation time
Tb = Backoff time

Advantages of CSMA:

1. **Increased efficiency:** CSMA ensures that only one device communicates on the network at a time, reducing collisions and improving network efficiency.
2. **Simplicity:** CSMA is a simple protocol that is easy to implement and does not require complex hardware or software.
3. **Flexibility:** CSMA is a flexible protocol that can be used in a wide range of network environments, including wired and wireless networks.
4. **Low cost:** CSMA does not require expensive hardware or software, making it a cost-effective solution for network communication.

Disadvantages of CSMA:

1.**Limited scalability:** CSMA is not a scalable protocol and can become inefficient as the number of devices on the network increases.

2.**Delay:** In busy networks, the requirement to sense the medium and wait for an available channel can result in delays and increased latency.

3.**Limited reliability:** CSMA can be affected by interference, noise, and other factors, resulting in unreliable communication.

4.**Vulnerability to attacks:** CSMA can be vulnerable to certain types of attacks, such as jamming and denial-of-service attacks, which can disrupt network communication.

B. Controlled Access Protocol

It is a method of reducing data frame collision on a shared channel. In the controlled access method, each station interacts and decides to send a data frame by a particular station approved by all other stations. It means that a single station cannot send the data frames unless all other stations are not approved. It has three types of controlled access: **Reservation, Polling**, and **Token Passing**.

C. Channelization Protocols

It is a channelization protocol that allows the total usable bandwidth in a shared channel to be shared across multiple stations based on their time, distance and codes. It can access all the stations at the same time to send the data frames to the channel.

Following are the various methods to access the channel based on their time, distance and codes:

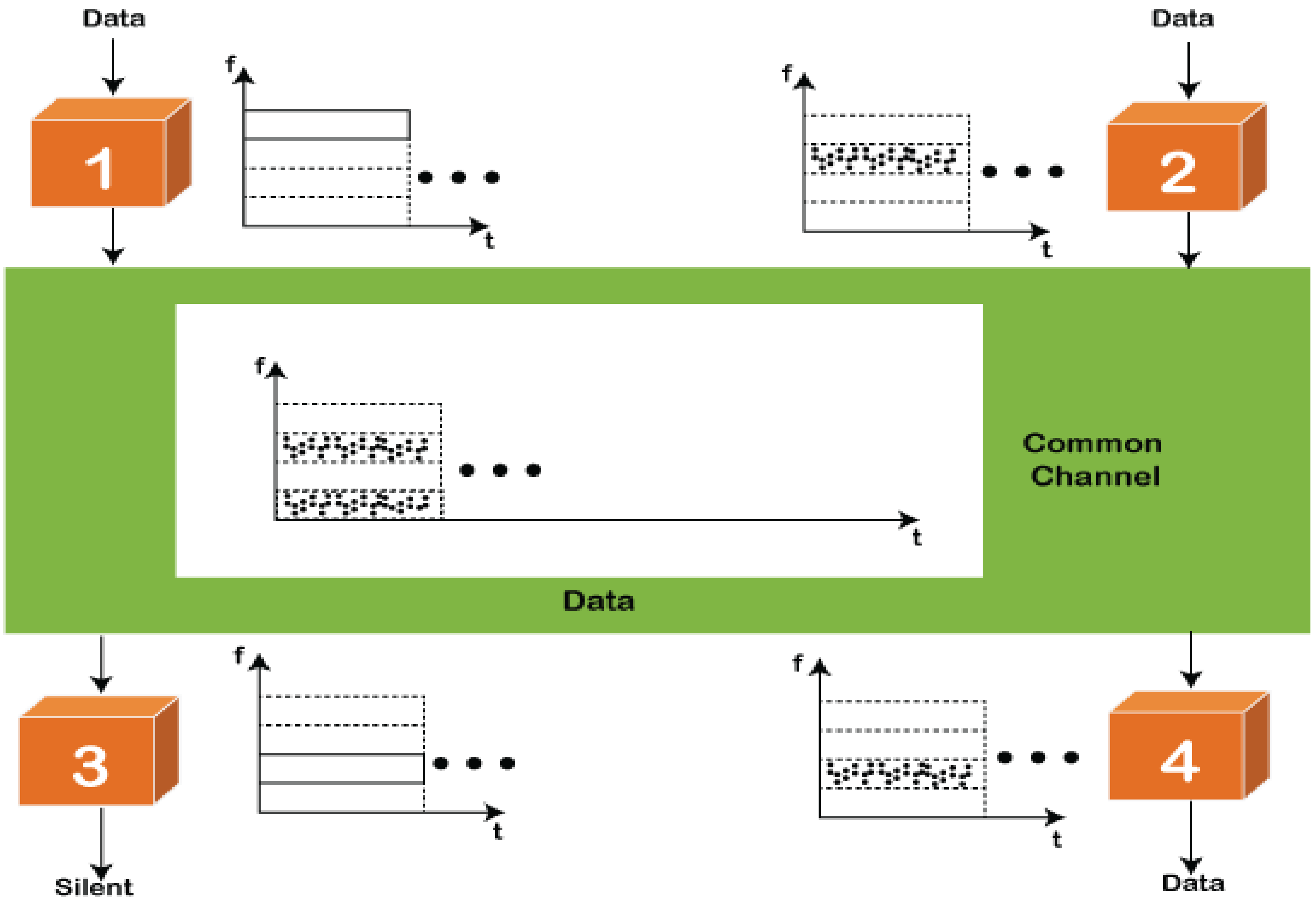
FDMA (Frequency Division Multiple Access)

TDMA (Time Division Multiple Access)

CDMA (Code Division Multiple Access)

FDMA

It is a frequency division multiple access (**FDMA**) method used to divide the available bandwidth into equal bands so that multiple users can send data through a different frequency to the subchannel. Each station is reserved with a particular band to prevent the crosstalk between the channels and interferences of stations.



TDMA

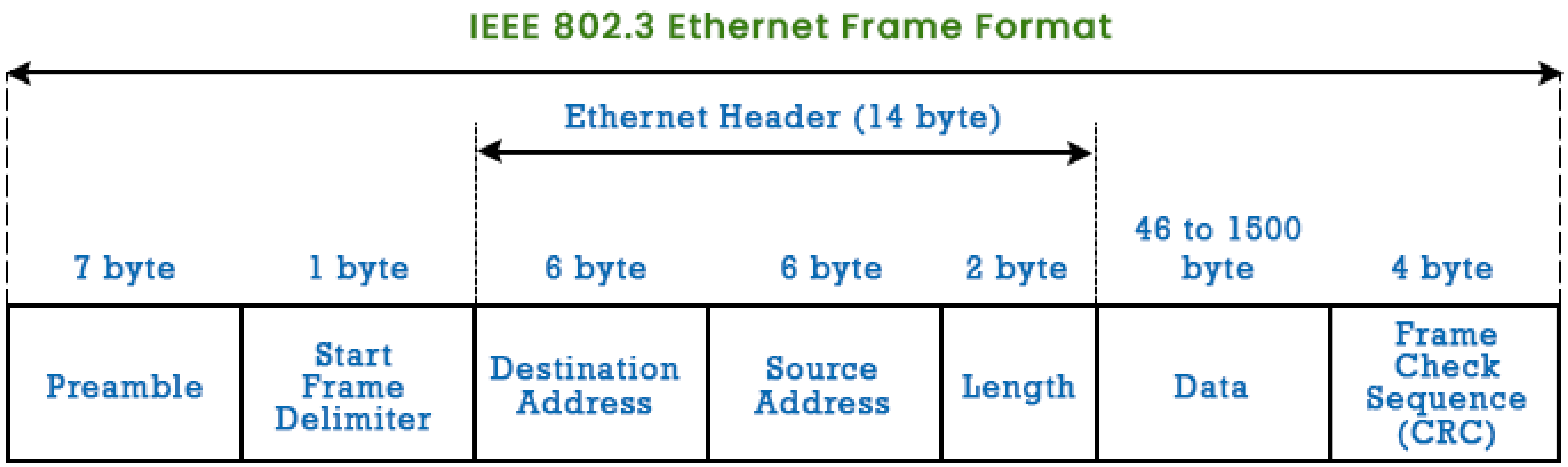
Time Division Multiple Access (**TDMA**) is a channel access method. It allows the same frequency bandwidth to be shared across multiple stations. And to avoid collisions in the shared channel, it divides the channel into different frequency slots that allocate stations to transmit the data frames. The same **frequency** bandwidth into the shared channel by dividing the signal into various time slots to transmit it. However, TDMA has an overhead of synchronization that specifies each station's time slot by adding synchronization bits to each slot.

CDMA

- The code division multiple access (CDMA) is a channel access method. In CDMA, all stations can simultaneously send the data over the same channel.
- It means that it allows each station to transmit the data frames with full frequency on the shared channel at all times.
- It does not require the division of bandwidth on a shared channel based on time slots. If multiple stations send data to a channel simultaneously, their data frames are separated by a unique code sequence.
- Each station has a different unique code for transmitting the data over a shared channel.
- For example, there are multiple users in a room that are continuously speaking. Data is received by the users if only two-person interact with each other using the same language. Similarly, in the network, if different stations communicate with each other simultaneously with different code language.

IEEE 802.3 Standards and Frame Formats

The IEEE 802.3 standard defines the fundamental frame format that is necessary for all MAC implementations. However, the core functionality of the protocol is being extended by several optional forms.



Preamble and SFD, which operate at the physical layer, begin an Ethernet frame. The packet's payload follows the Ethernet header, which includes the MAC addresses for the source and destination. CRC, the final field, is utilized to find errors.

Let's now examine each section of the fundamental frame format.

1. **PREAMBLE** - Ethernet frames begin with a 7-byte. This is a sequence of alternate 0s and 1s that denotes the beginning of the frame and enables bit synchronization between the sender and receiver. PRE (Preamble) was initially developed to accommodate the loss of a few bits as a result of signal delays. However, the frame bits in high-speed Ethernet today are protected without the need for a preamble. Prior to the actual frame beginning, PRE (Preamble) alerts the receiver that a frame is about to start and enables the receiver to lock onto the data stream.

2. **Start of frame Delimiter (SFD)** - This 1-byte field is always set to **10101011**. The destination address is the next set of bits that will begin the frame, as indicated by SFD. The preamble is frequently referred to as 8 Bytes since SFD is sometimes seen as a component of PRE. The SFD notifies the station or stations that synchronization is now impossible.

3. **Destination Address** - This 6-Byte element contains the MAC address of the device for which the data is intended.

1. **Source Address** - This 6-byte element contains the source machine's MAC address. Since Source Address is always a unique address (Unicast), 0 is always the least significant bit of the first byte.
2. **Length** - A 2-Byte field called Length represents the size of an Ethernet frame as a whole. Due to some inherent constraints of Ethernet, this 16-bit field can store length values from 0 to 65534, but length values greater than 1500 are not permitted.
3. **Data** - This area, sometimes referred to as the Payload, is where the real data is placed. If Internet Protocol is utilised via Ethernet, both the IP header and data will be placed here. The longest possible piece of data might be 1500 bytes long. If the data length is less than the minimum length, which is 46 bytes, padding 0's are appended to make up the difference.
4. **Cyclic Redundancy Check (CRC)** - CRC is a field of 4 bytes. The data in this field is a 32-bit hash code created using the fields for the destination address, source address, length, and data. Data is damaged if the checksum calculated by the destination differs from the checksum value supplied.

DA [Destination MAC Address]: ***6 bytes***

SA [Source MAC Address]: ***6 bytes***

Type [0x8870 (Ethertype)]: ***2 bytes***

DSAP [802.2 Destination Service Access Point]: ***1 byte***

SSAP [802.2 Source Service Access Point]: ***1 byte***

Ctrl [802.2 Control Field]: ***1 byte***

Data [Protocol Data]: > ***46 bytes***

FCS [Frame Checksum]: ***4 bytes***

Advantages of using Ethernet:

1.Simple to implement

2.Maintenance is Easy

3.Less cost

What is Fast Ethernet?

- In the fast-evolving world of computers, speed plays a vital role in the present and has become paramount. As the technology is advancing and the demands on the networks are increasing significantly, the quest for transferring the data at faster rates has become a priority.
- A very crucial evolution in networking technology developed to meet the escalating features and demands head-on – enters the Fast Ethernet.
- Fast Ethernet has transformed the landscape of communication enabling unparalleled speeds and unleashing a new era of connectivity through speed.

Primary Terminologies Related to Fast Ethernet

•**Ethernet:** Ethernet is a family of wired computers and It is a widely used networking technology that mainly defines the path of the data transmission over a wired connection. It works on the principle of packet switching and uses the ethernet protocol. Ethernet technology allows network-connected devices to communicate without packet collisions. In an Ethernet network, data is broken into packets.

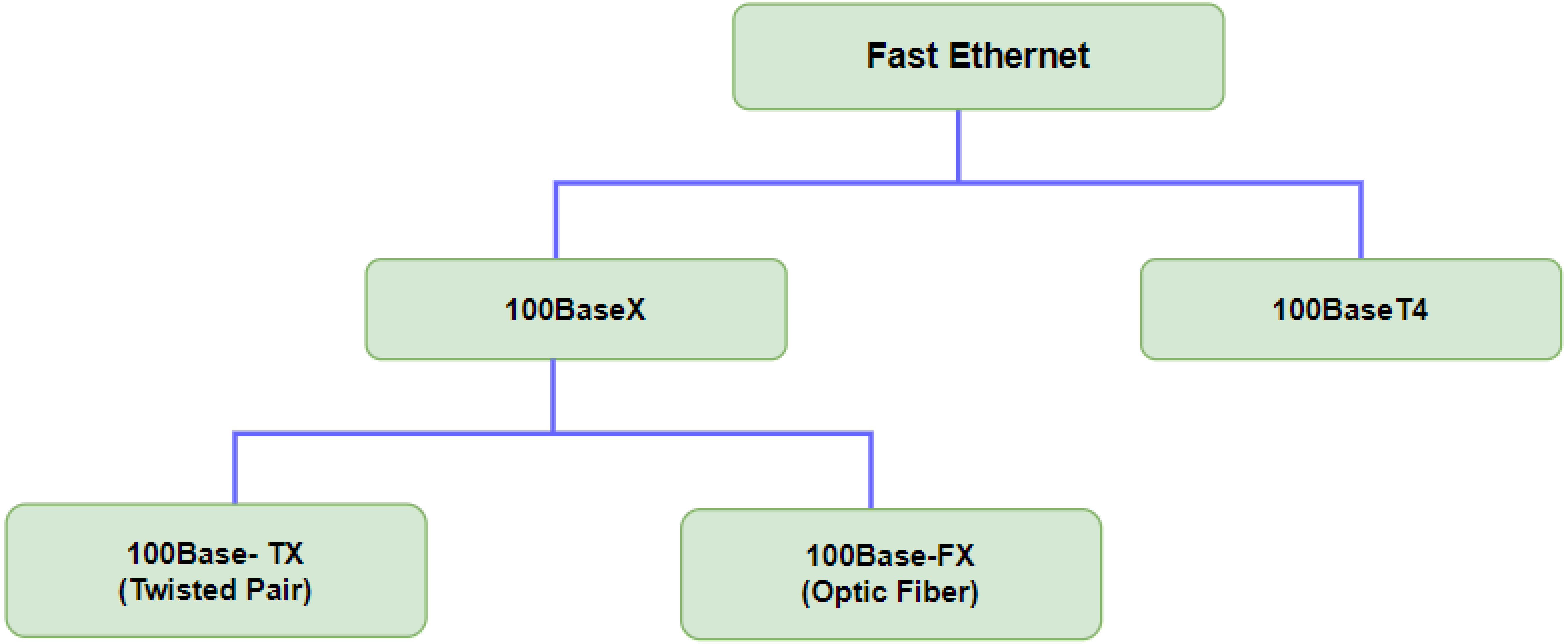
•**Data Transfer Rate (DTR):** Data Transfer rate also known as Data Throughput. It is the speed at which the data is transmitted from one device to another within the connected network. DTR is mainly measured in bits per second(bps) or in major kilobits per second(Kbps).

•**Band Width:** It mainly refers to the maximum rate of data transfer across a connected network medium. These represent the capacity of the medium which can carry data up to what extent. Bandwidth is actually the volume of information that can be sent over a connection in a measured amount of time – calculated in megabits per second (Mbps).

Types of Fast Ethernet

Fast Ethernet is mainly of several types or standards, each having their own specifications, implementations and characteristics. The two most common types of Fast Ethernet are:

- 100Base-TX
- 100Base-FX



1. 100Base-TX

•100Base-TX utilizes twisted pair copper cabling specified as Cat5e cables for the ease of data transmission.

•It is mainly used in Local Area Networks(LANs)

•It works on both Full duplex and Half Duplex Modes and it supports data transfer rates up to 100 Mbps.

•This is used in the connecting devices like computers. printers and LAN environment.

2. 100Base-FX

•100Base-FX is another type of Fast Ethernet that is in different cabling which is fiber optic cabling for Data transmission

•It is mainly used in long run cables of range 120 Kms and it requires electro magnetic interference.

•It supports up to 100 Mbps of DTR and it also offers the higher bandwidth.

•This is used commonly in connecting devices across different buildings or in the environment where copper cabling is out of reach.

Significance of Fast Ethernet

The Significance of Fast Ethernet lies in the impact on networking capabilities, performance and Efficiency.

Some Key points mentioned are the significance of Fast Ethernet below:

- Increased Data Transfer Rates
- Enhanced Network Performance
- Cost Effectiveness
- Scalability
- Flexibility
- Support for Bandwidth intensive applications (Video Conferencing, Multimedia streaming, Large File Transfers).

Applications of Fast Ethernet

The applications of Fast Ethernet are diverse in various industries and scenarios. The following are some of the applications for Fast Ethernet:

•**Data Centers:** Fast Ethernet is commonly deployed in data center environments to connect servers, storage devices, and networking equipment. It facilitates rapid data transfer between servers and storage arrays, supporting mission-critical applications and services hosted in the data center.

•**Surveillance Systems:** Fast Ethernet is utilized in surveillance systems for transmitting high-definition video feeds from IP cameras to monitoring stations or recording devices. It ensures real-time monitoring and recording of surveillance footage, enhancing security and surveillance capabilities.

- **Educational Institutions:** Fast Ethernet is employed in educational institutions, such as schools and universities, to support networked learning environments. It enables access to online educational resources, collaborative tools, and e-learning platforms, enriching the educational experience for students and educators.
- **Telecommunications:** Fast Ethernet is used in telecommunications networks for backhaul connections between central offices, cell towers, and network aggregation points. It enables the efficient transfer of voice, data, and video traffic over large-scale telecommunications networks.

Introduction of Gigabit Ethernet

•The committee began working on a faster Ethernet, quickly dubbed gigabit Ethernet. The goal was to increase performance while maintaining all Ethernet standards. Gigabit Ethernet had to provide service with both unicast and broadcast using the same 48-bit address scheme and also maintaining the same frame format. All configurations of gigabit Ethernet must use point-to-point links. In fig.(a) two computers are directly attached to each other.

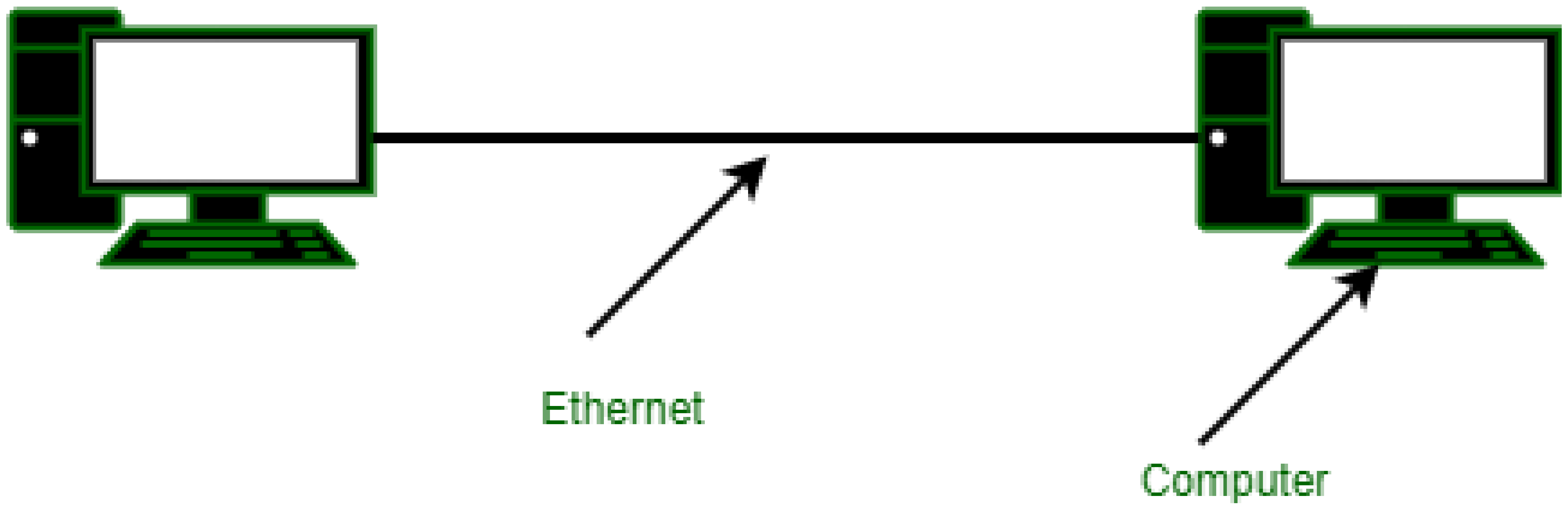


fig.(a) A two station Ethernet

However, uses a switch or hub connected to multiple computers and possibly additional switches or hubs as shown in fig.(b). In both figures, the Ethernet cable has exactly two devices on it.

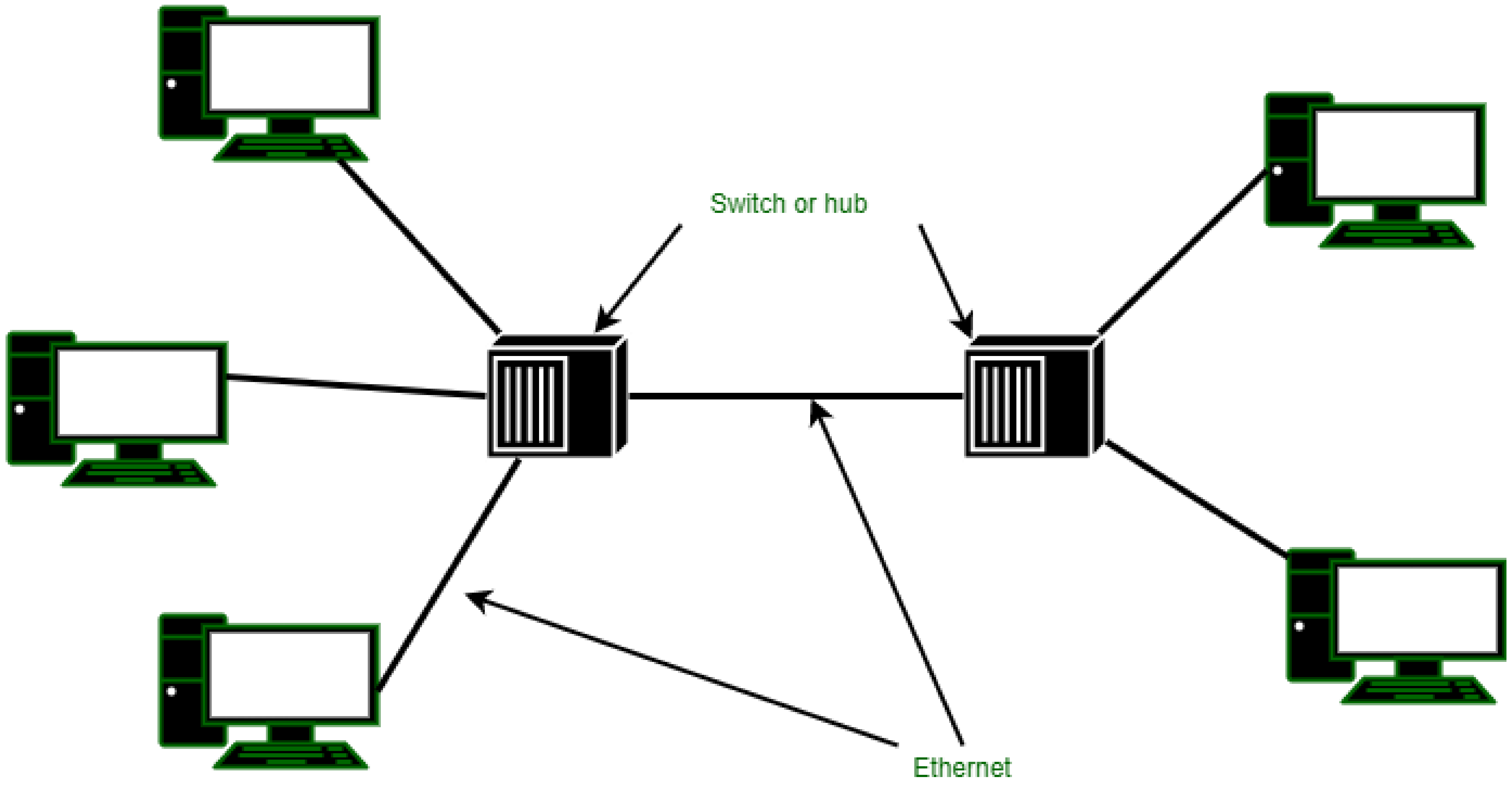


fig.(b) A multistation Ethernet

Features of Gigabit Ethernet :

- It supports two different modes i.e. full-duplex mode and half-duplex mode. Full-duplex mode allows traffic in two directions at the same time. When a central switch connected to computers on the periphery this mode is used.
- In this, all lines are buffered so each computer and switch is free to send frames whenever it wants to. In this mode, contention is not possible.
- The computer is the only possible sender to the switch, and transmission will succeed even if the switch is currently sending a frame to the computer.
- A half-duplex mode is used when computers are connected to a hub rather than a switch. A hub does not buffer incoming frames.
- All the lines are electrically connected internally, simulating the multi-drop cable used in classic Ethernet. Standard CSMA/CD protocol is required in this mode because collisions are possible. Because a 64-byte frame can now be transmitted 100 times faster than in classic Ethernet, the maximum cable length must be 100 times less or 25 meters.

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Advantages of Gigabit Ethernet:

1. You can acquire the services with low cost.
2. Easily large amount of data can be transferred across network.
3. It offers full-duplex capacity, so bandwidth can be virtually doubled.
4. It reduces bottleneck problems and enhances performances.

Disadvantages of Gigabit Ethernet:

1. Full duplex mode is not supported in 1000Base-T
2. 4 pairs of wiring are required in 1000Base-T to transfer data.

IEEE 802.11 Architecture

- The IEEE 802.11 standard, commonly known as Wi-Fi, outlines the architecture and defines the MAC and physical layer specifications for wireless LANs (WLANs). Wi-Fi uses high-frequency radio waves instead of cables for connecting the devices in LAN.
- Given the mobility of WLAN nodes, they can move unrestricted within the network coverage zone. The 802.11 structure is designed to accommodate mobile stations that participate actively in network decisions. Furthermore, it can seamlessly integrate with 2G, 3G, and 4G networks.
- The Wi-Fi standard represents a set of wireless LAN standards developed by the Working Group of IEEE LAN/MAN standards committee (IEEE 802). The term 802.11x is also used to denote the set of standards. Various specifications and amendments include 802.11a, 802.11b, 802.11e, 802.11g, 802.11n etc.

Frame Format of IEEE 802.11

IEEE 802.11 MAC layer data frame consists of 9 fields:

Frame control	Duration /ID	Address 1	Address 2	Address 3	SC	Address 4	Data	CRC
2 bytes	2 bytes	6 bytes	6 bytes	6 bytes	2 bytes	6 bytes	0 - 2312 bytes	4 bytes

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

IEEE 802.11 MAC Frame Structure

Frame Control

It is 2 bytes long and defines type of frame and control information. The types of fields present in FC are:

- **Version:** Indicates the current protocol version.
- **Type**: Determines the function of frame i.e. management(00), control(01) or data(10).
- **Subtype**: Indicates subtype of frame like 0000 for association request, 1000 for beacon.
- **To DS**: When set indicates that the destination frame is for DS(distribution system).
- **From DS:** When set indicates frame coming from DS.
- **More frag (More fragments)**: When set to 1 means frame is followed by other fragments.
- **Retry: I**f the current frame is a re-transmission of an earlier frame, this bit is set to 1.

- **Power Mgmt (Power Management):** It indicates the mode of a station after successful transmission of a frame. Set to ‘1’ field indicates that the station goes into power-save mode. If the field is set to 0, the station stays active.
- **More data:** It is used to indicate to the receiver that a sender has more data to send than the current frame.
- **WEP**: It indicates that the standard security mechanism of 802.11 is applied.
- **Order:** If this bit is set to 1 the received frames must be processed in strict order.

Advantages and Disadvantages of IEEE 802.11 Architecture

There are some list of Advantages and Disadvantages of IEEE 802.11 Architecture are given below :

Advantages of IEEE 802.11 Architecture

- **Fault Tolerance:** The centralized architecture minimizes the bottlenecks and introduces resilience in the WLAN equipment.
- **Flexible Architecture**: Supports both temporary smaller networks and larger, more permanent ones.
- **Prolonged Battery Life:** Efficient power-saving protocols extend mobile device battery life without compromising network connections.

Disadvantages of IEEE 802.11 Architecture

- **Noisy Channels:** Due to reliance on radio waves, signals may experience interference from nearby devices.
- **Greater Bandwidth and Complexity:** Due to necessary data encryption and susceptibility to errors, WLANs need more bandwidth than their wired counterparts.
- **Speed:** Generally, WLANs offer slower speeds compared to wired LANs.